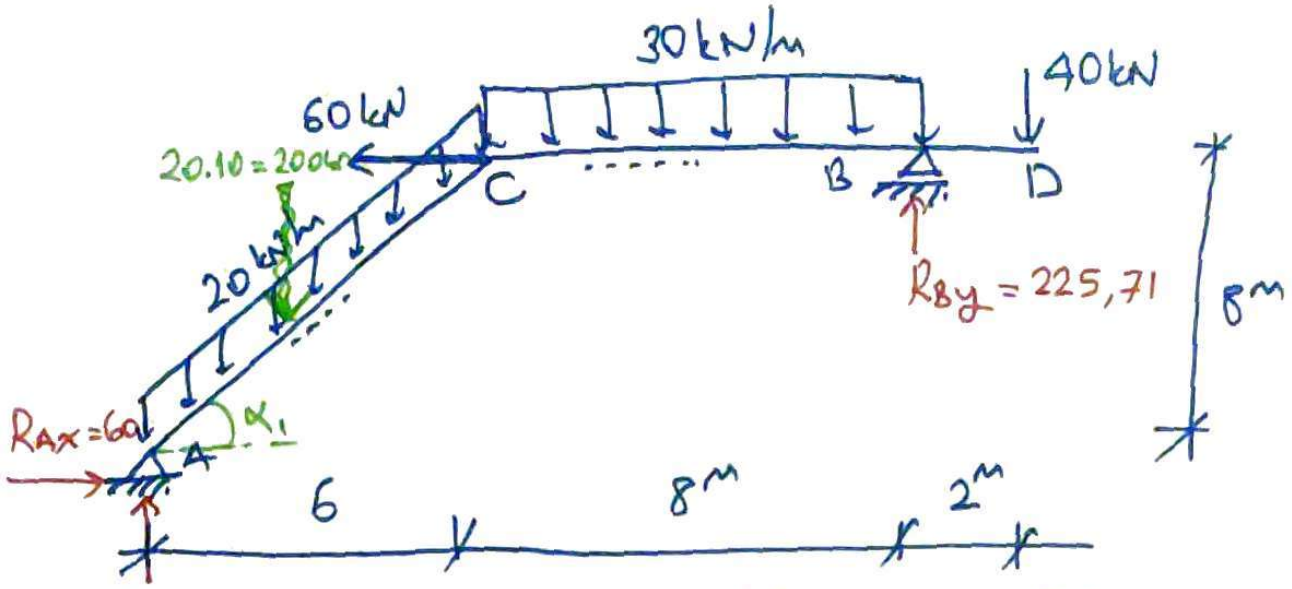
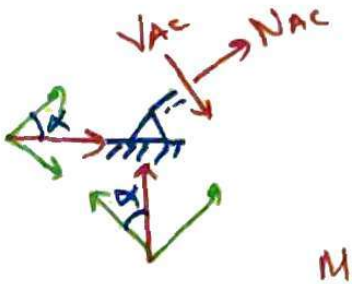


$\sin\alpha = 0,80$; $\cos\alpha = 0,60$

$\Sigma F_x = 0 \Rightarrow R_{Ax} = 60\,kN$

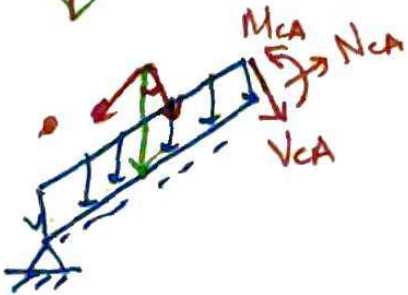
$\Sigma M_B = 0 \Rightarrow R_{Ay} = \dfrac{20.10.11 + 30.8.4 - 40.2 + 60.8}{14}\,kN = 254,29$

$\Sigma M_A = 0 \Rightarrow R_{By} = \dfrac{40.16 + 30.8.10 + 20.10.3 - 60.8}{14} = 225,71\,kN$



$V_{AC} = -60.\sin\alpha + 254,29.\cos\alpha = 104,57\,kN$

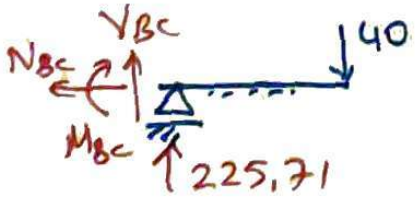
$N_{AC} = -60.\cos\alpha - 254,29.\sin\alpha = -239,43\,kN$



$V_{CA} = -60.\sin\alpha + 254,29\cos\alpha - 200.\cos\alpha$
$= -15,43\,kN$

$N_{CA} = -60.\cos\alpha - 254,29.\sin\alpha + 200\sin\alpha$
$= -79,43$

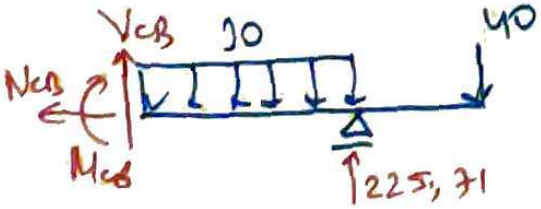
$M_{CA} = -60.8 + 254,29.6 - 200.3 = 445,74\,kNm$



$V_{BC} = -225,71 + 40 = -185,71\,kN$

$N_{BC} = 0$

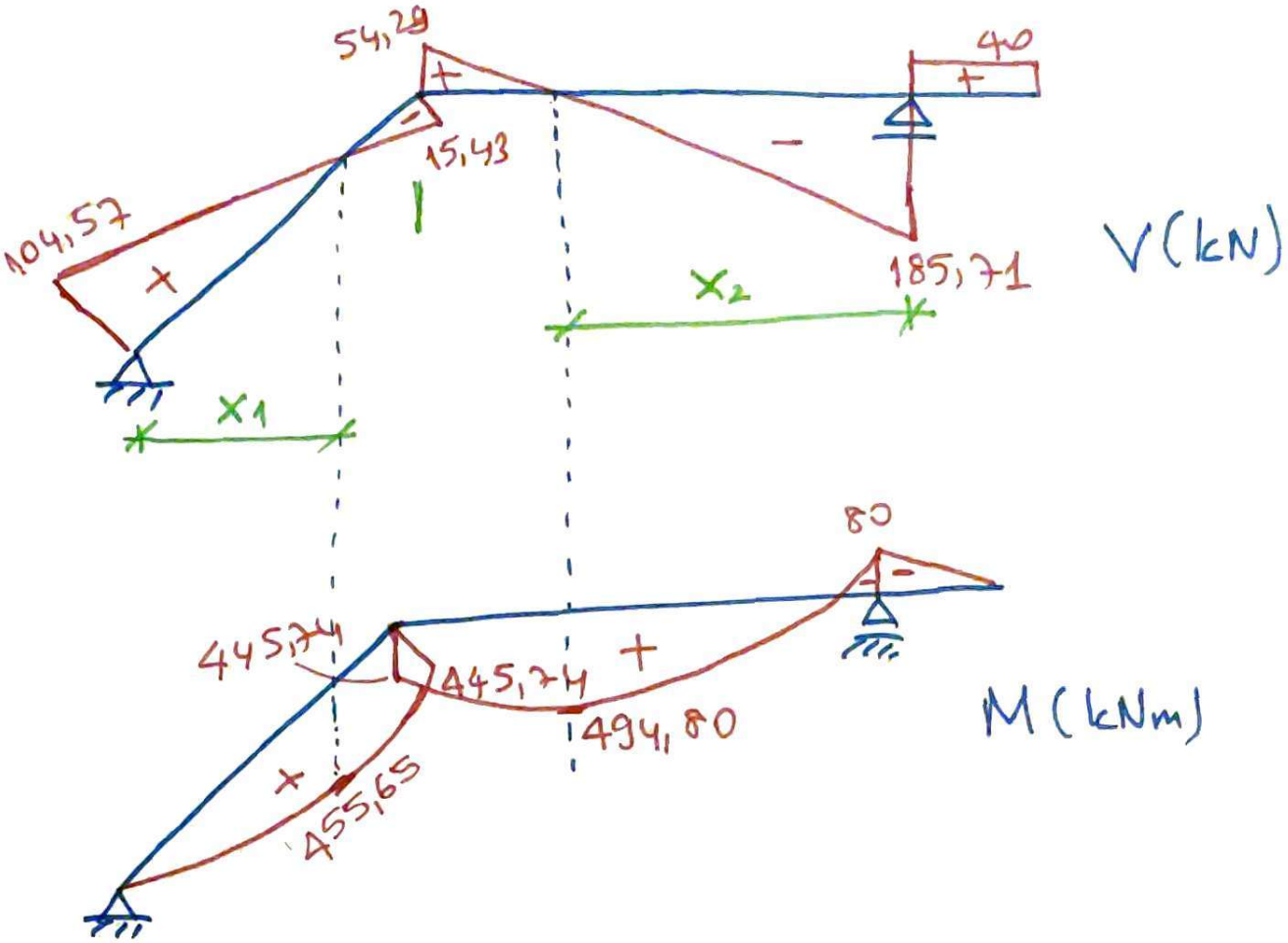
$M_{BC} = M_{BD} = -40.2 = -80\,kNm$

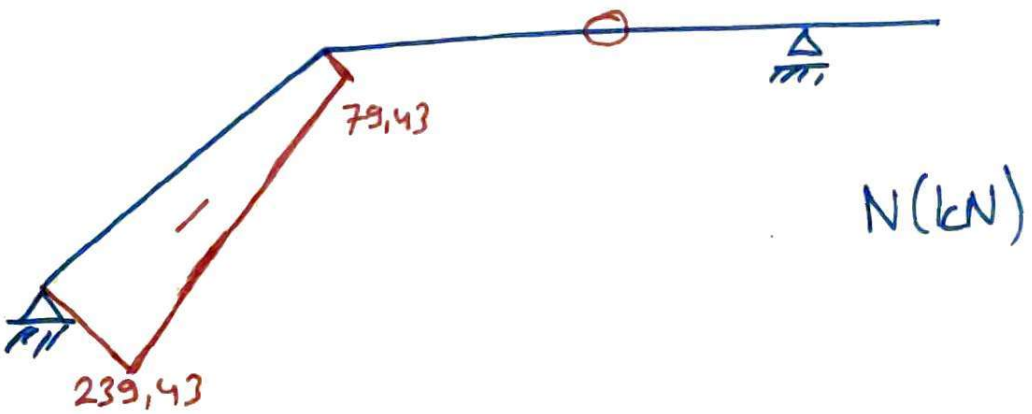


$V_{CB} = 30.8 + 40 - 225,71 = 54,29\,kN$

$N_{CB} = 0$

$M_{CB} = 225,71.8 - 40.10 - 30.8.4 = 445,68\,kNm$





$$20.\frac{x_1}{\cos\alpha}.\cos\alpha + 60.\sin\alpha - 254,29.\cos\alpha = 0$$

$$x_1 = 5,23\ m$$

$$M_{max,Ac} = 254,29.5,23 - 20.\frac{5,23}{\cos\alpha}.\frac{5,23}{2} - 60.5,23.\frac{\sin\alpha}{\cos\alpha}$$

$$= 455,65\ kNm$$

$$30.x_2 = 225,71 - 40 \Rightarrow x_2 = 6,19\ m$$

$$M_{max,Bc} = 225,71.6,19 - 30.\frac{6,19^2}{2} - 40.8,19$$

$$= 494,80\ kNm$$